

Real-Life + Fun Simulation Problems

All problems focus on:

- function behavior
 - print vs return
 - None checking
 - if/else flow
 - logic building
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1. Candy Machine Simulator

A shop sells one candy for 10 rupees.

Write a function `buy_candy(money)` that:

- If the student gives 10 or more rupees → return `"Candy bought!"`
- If the money is less than 10 → print `"Not enough money"`

After calling the function:

- If the function returned something, print: `"Customer got the candy"`
 - If nothing returned, print: `"No candy given"`
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2. PUBG/FreeFire Health Booster

A player's health is from 0 to 100.

Create a function `heal(health)` :

- Print `"Healing..."` every time the function runs
- If health is 100 or more → return nothing (returns None)
- Else → return `"Healed"`

After calling:

- If a value was returned → print `"Healing successful"`
 - If it returned None → print `"Player already at full health"`
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3. ATM Withdrawal Machine

Write a function `withdraw(balance, amount)` :

- If amount is greater than balance → print `"Insufficient balance"`
- Otherwise → return the new balance (balance - amount)

After calling:

- If it returned a value, print `"New balance: <value>"`
 - If it didn't return a value, print `"Transaction failed"`
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4. School Attendance System

Create a function `mark_attendance(name, present)` :

- If `present` is True → return `"Marked Present"`
- If `present` is False → print `"Student is absent"`

After calling:

- If return value exists → print `"Attendance saved"`
 - Else → print `"Attendance not saved"`
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5. Mobile Battery Checker

Create a function `battery_status(level)` :

- If battery level is less than 20 → print `"Low battery"`
- Otherwise return `"Battery OK"`

After calling:

- Print `"Phone is functioning normally"` if a value is returned
 - Otherwise print `"Phone battery warning"`
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6. Restaurant Billing System

Create a function `calculate_bill(items)`

(Here, items is a list of item prices, for example `[100, 50, 130]`)

The function should:

- If the list is empty → print `"No items selected"`
- Otherwise → return the total sum

After calling:

- If value returned → print `"Total bill is: <value>"`
 - Otherwise → print `"Cannot generate bill"`
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7. Weather Alert System

Create a function `check_temp(temp)` :

- If `temp > 40` → print `"Too hot! Stay hydrated"`
- If `temp < 0` → print `"Freezing cold! Dress warmly"`
- Otherwise → return `"Weather is normal"`

Then check if the function returned a value and display it.

8. Game XP Reward System

Create a function `reward(xp)` :

- If xp is greater than 1000 → return `"Level Up!"`
- Otherwise → print `"Earn more XP to level up"`

After calling:

- If return value exists → print "Congratulations!"
 - Otherwise → print "Try again"
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9. Delivery Service Simulator

A company only delivers within 50 km.

Write function `delivery_status(distance)` :

- If distance > 50 → print "Delivery not available"
- Else → return "Delivery on the way"

After calling:

- If return value exists → print "Order placed successfully"
 - Else → print "Order failed"
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10. Traffic Light System

Create a function `traffic(color)` :

- If color is "red" → print "Stop"
- If "yellow" → print "Get ready"
- If "green" → return "Go"

After calling:

- If value returned → print "Drive"
 - Otherwise → print "Wait"
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11. YouTube Video Loader

Create a function `load_video(speed)` :

- If speed < 2 Mbps → print "Buffering..."
- Else → return "Playing Video"

After calling:

- If returned → print `"Enjoy your video!"`
 - Else → print `"Internet slow, please wait"`
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12. Grocery Checkout

Create a function `checkout(amount)` :

- If amount is 0 → print `"Your cart is empty"`
- Else → return `"Payment successful"`

After calling:

- If returned → print `"Thank you for shopping!"`
 - Else → print `"Could not checkout"`
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13. Password Strength Checker

Create a function `check_password(password)` :

- If password length is less than 8 → print `"Weak password"`
- Else → return `"Strong password"`

After calling, print:

- `"Secure"` if returned value exists
 - `"Please choose stronger password"` otherwise
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14. Bus Booking System

Create a function `book_ticket(seats)` :

- If seats == 0 → print `"Sold out"`
- Else → return `"Ticket booked"`

After calling, check if the function returned something.

15. Nepali Mobile Recharge Simulator

Create a function `recharge(amount)` :

- If amount < 10 → print `"Minimum recharge is 10"`
- Else → return `"Recharge successful"`

After calling:

- If returned → print `"Balance Updated"`
 - Else → print `"Recharge failed"`
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16. Water Bottle Level Checker

Create a function `check_water(level)` :

- If level == 0 → print `"Bottle empty"`
- If level < 50 → print `"Need refill soon"`
- If level >= 50 → return `"Water level OK"`

Use return checking.

17. Movie Ticket Counter

Function `buy_ticket(age)` :

- If age < 5 → print `"Free entry"`
- If age between 5-17 → return `"Child ticket"`
- If age 18 or more → return `"Adult ticket"`

After calling, check if returned or only printed.

18. Laptop Battery Saver Mode

Function `power_mode(level)` :

- If `level < 20` → print `"Switching to battery saver"`
 - Else → return `"Normal mode"`
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19. Fitness App Calories Counter

Function `calories_burned(minutes)` :

- If `minutes == 0` → print `"Workout not started"`
- Else → return `minutes * 5`

Check if returned.

20. Coffee Vending Machine

Function `make_coffee(coins)` :

- Coffee costs 15 rupees
 - If `coins < 15` → print `"Not enough coins"`
 - Else → return `"Coffee ready"`
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